Why Study Excel?

Excel is the world's most used spreadsheet program.

Example use areas:

1. Data analytics
2. Project management
3. Finance and accounting
4. It is continuously supported by Microsoft
5. Templates and frameworks can be reused by yourself and others, lowering creation costs

What is Excel[1]?

Excel is pronounced "Eks - sel"

It is a spreadsheet program developed by Microsoft. Excel organizes data in columns and rows and allows you to do mathematical functions. It runs on Windows, macOS, Android and iOS.

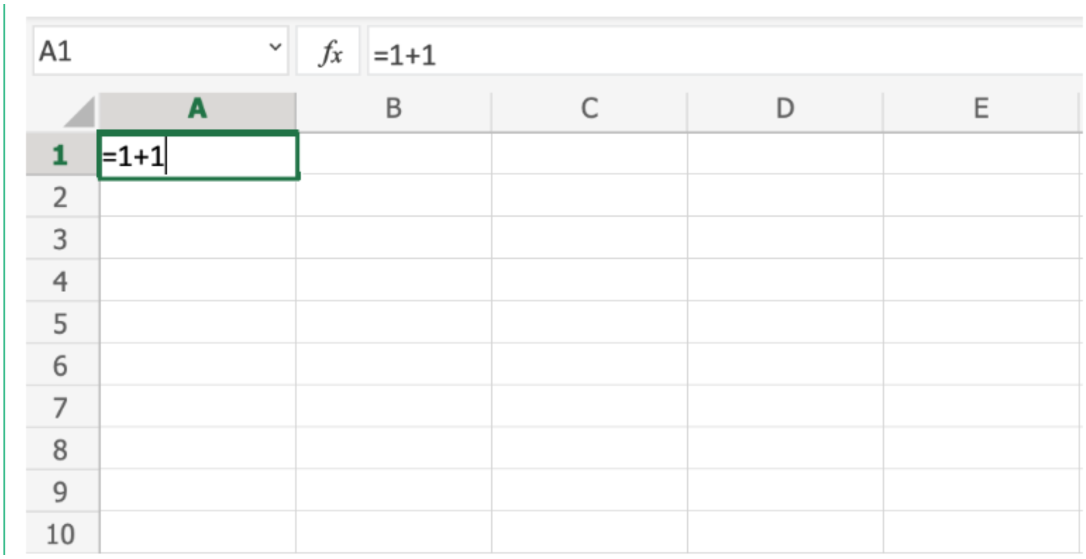
The first version was released in 1985 and has gone through several changes over the years. However, the main functionality mostly remains the same.

Excel is typically used for:

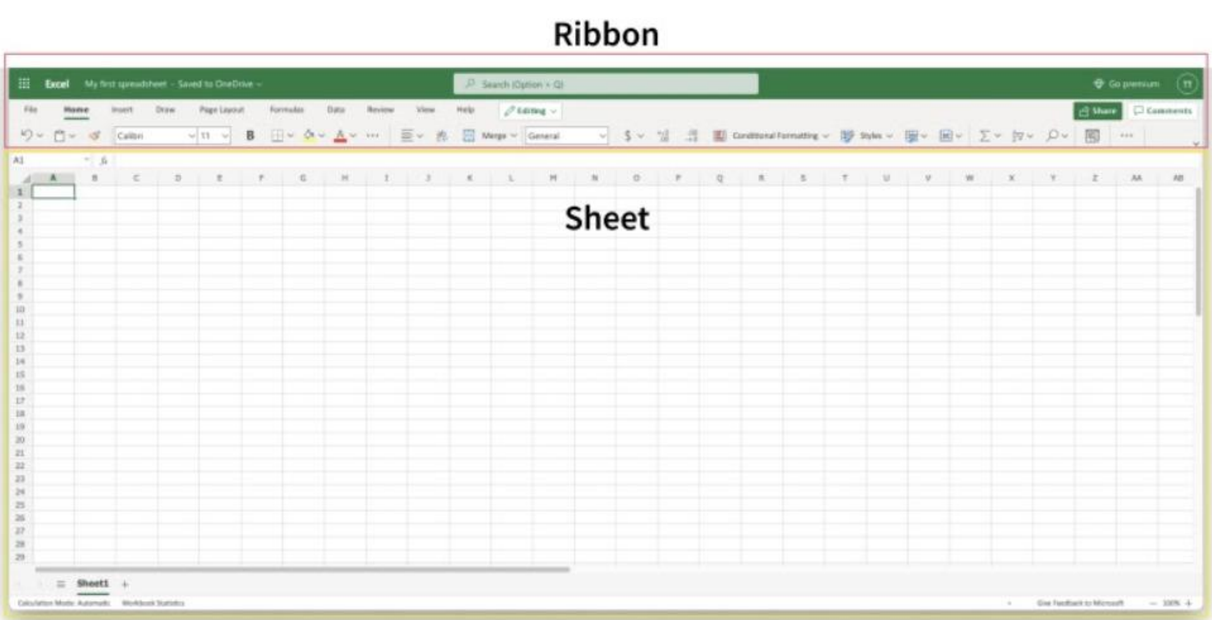
- Analysis
- Data entry
- Data management
- Accounting
- Budgeting
- Data analysis
- Visuals and graphs
- Programming
- Financial modeling
- And much, much more!

[1] https://www.office.com/

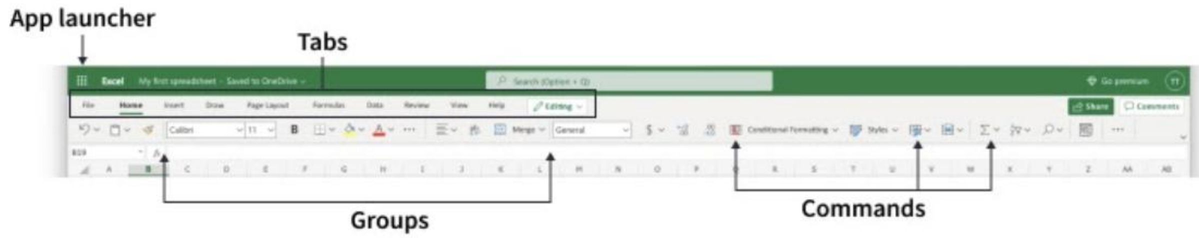
- First, double click the cell A1, the one that is marked with the green rectangle in the picture.
- Second, type =1+1.
- Third, hit the **enter** button:



Excel's structure is made of two pieces, the **Ribbon** and the **Sheet**.

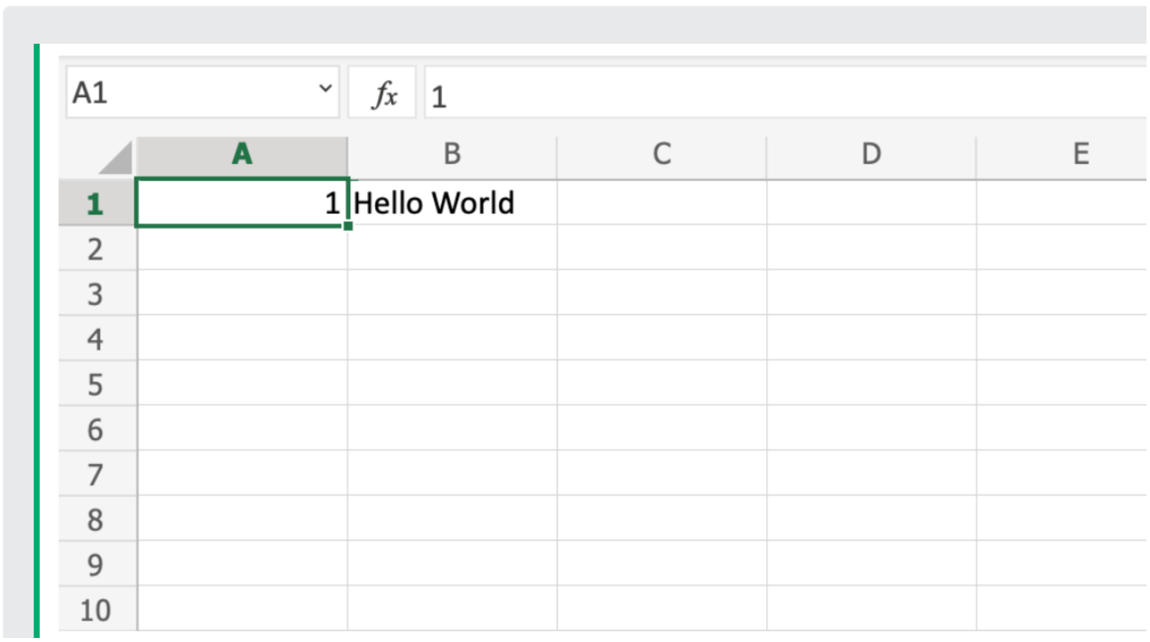


The **Ribbon** is made up by the **App launcher**, **Tabs**, **Groups** and **Commands**. In this section we will explain the different parts of the **Ribbon**.

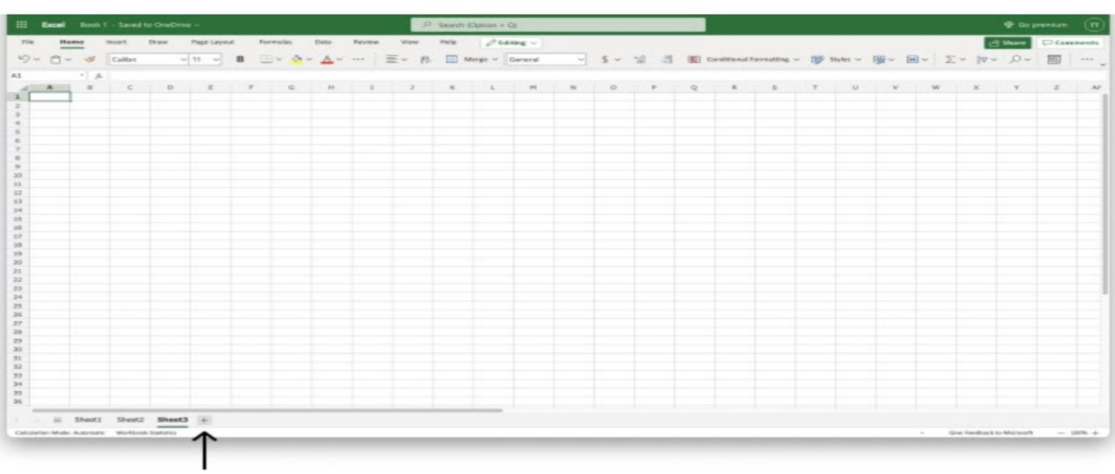


The **Sheet** is a set of rows and columns. It forms the same pattern as we have in math exercise books, the rectangle boxes formed by the pattern are called cells.

Values can be both numbers and letters:



Multiple Sheets (you can create new sheets)



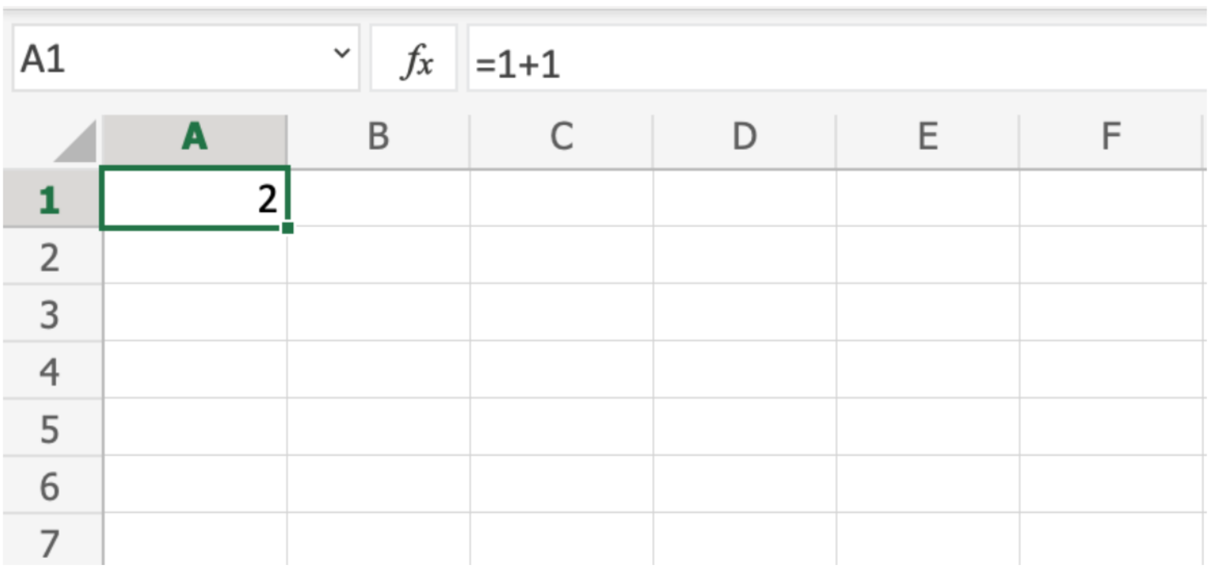
Syntax

A formula in Excel is used to do mathematical calculations. Formulas always start with the equal sign = typed in the cell, followed by your calculation.

Creating formulas, step by step

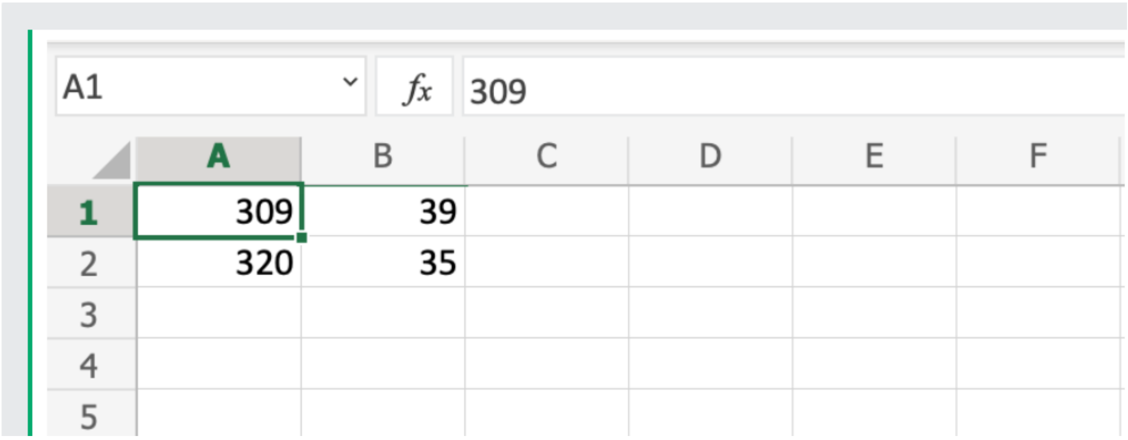
- Select a cell
- Type the equal sign (=)
- Select a cell or type value
- Enter an arithmetic operator
- Select another cell or type value
- Press enter.

For example, =1+1 is the formula to calculate **1+1=2**



Let's type some dummy values to get started. Double clicks the cells to type values into them. Go ahead and type:

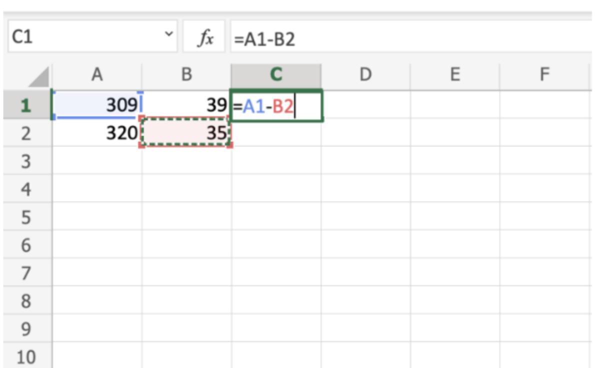
- A1(309)
- A2(320)
- B1(39)
- B2(35)

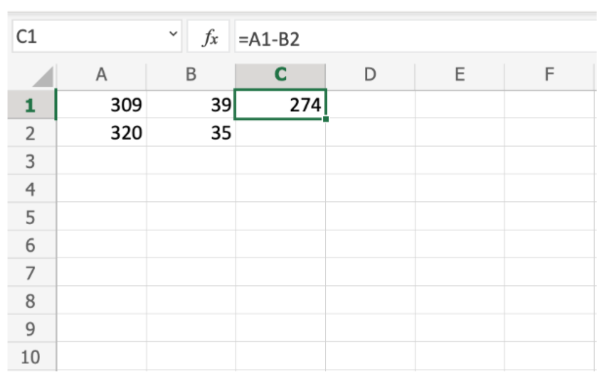


The formula can be typed directly without clicking the cells. The typed formula would be the same as the value in C1 (=A1-B2)

Here is how to do it, step by step.

1. Select the cell C1
2. Type the equal sign (=)
3. Left click on A1, the cell that has the (309) value
4. Type the minus sign (-)
5. Left click on B2, the cell that has the (35) value
6. Hit enter





To do

1- A1-B2

2- Apply Arithmetic Operation

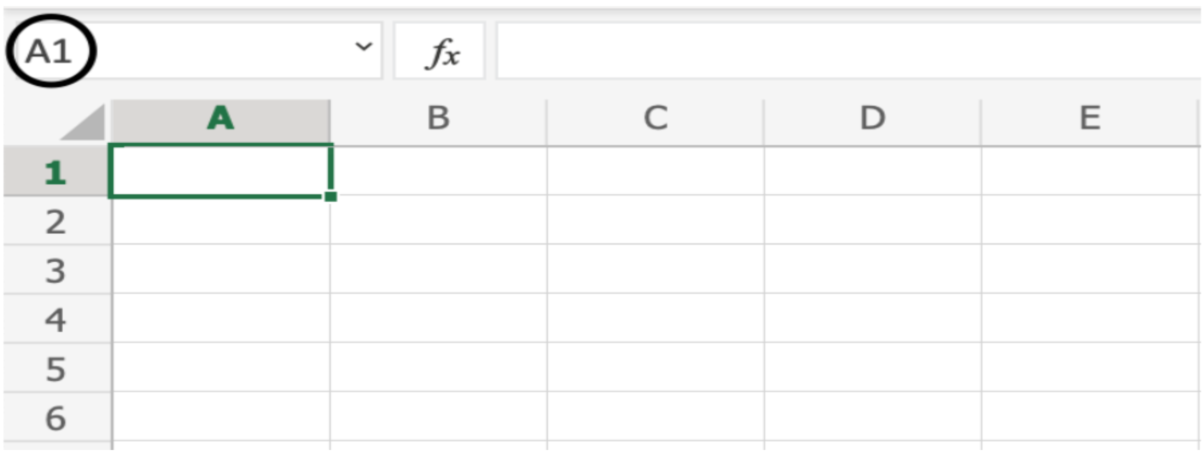
Excel Ranges

Range is an important part of Excel because it allows you to work with selections of cells.

There are four different operations for selection;

- Selecting a cell
- Selecting multiple cells
- Selecting a column
- Selecting a row

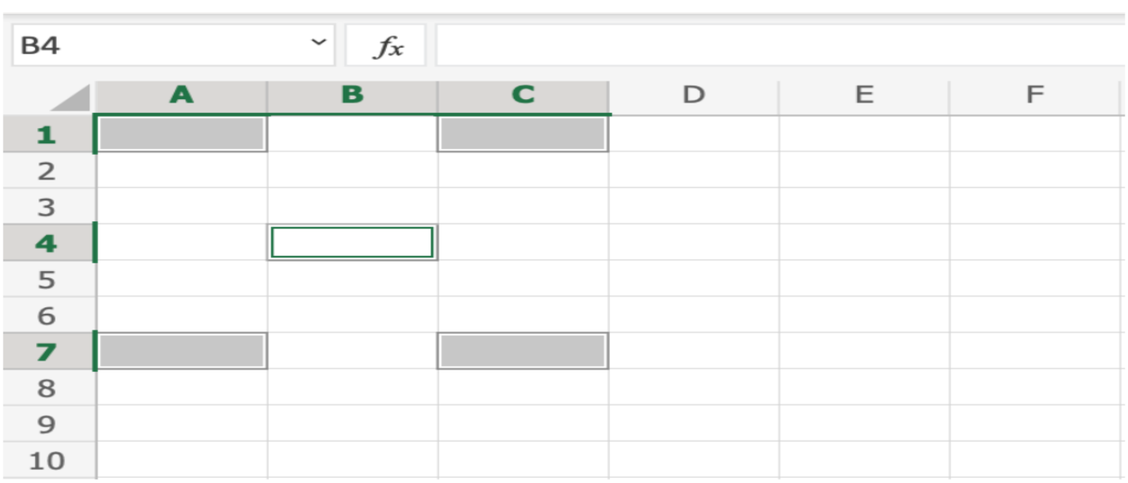
We introduce the Name Box as you can see in the top left.



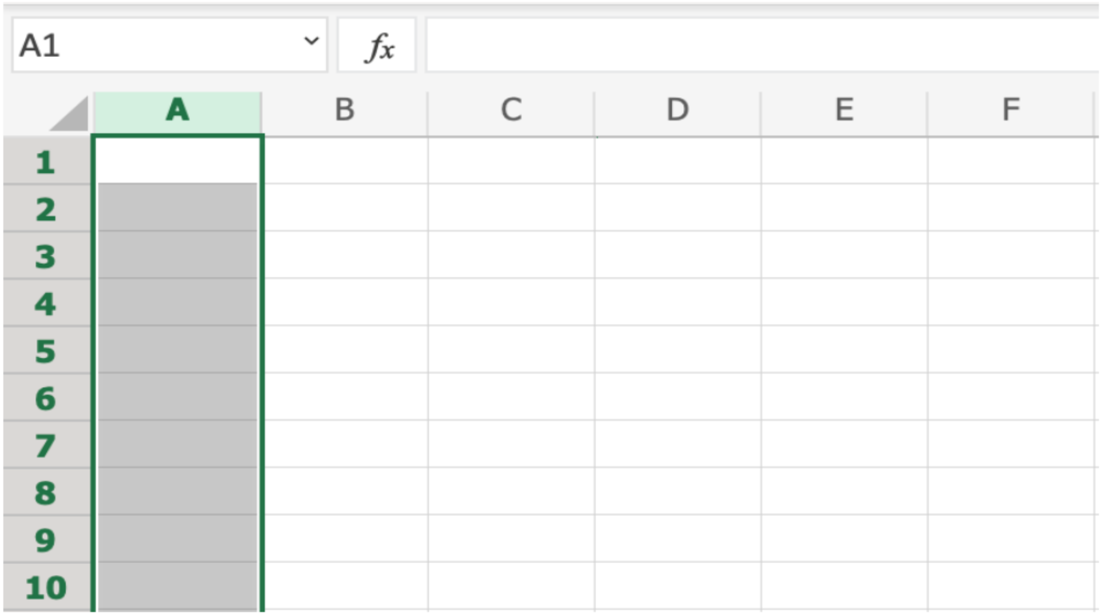
Selecting Multiple Cells

More than one cell can be selected by pressing and holding down <u>**CTRL**</u> and ***left*** clicking the cells. Once finished with selecting, you can let go of **CTRL**.

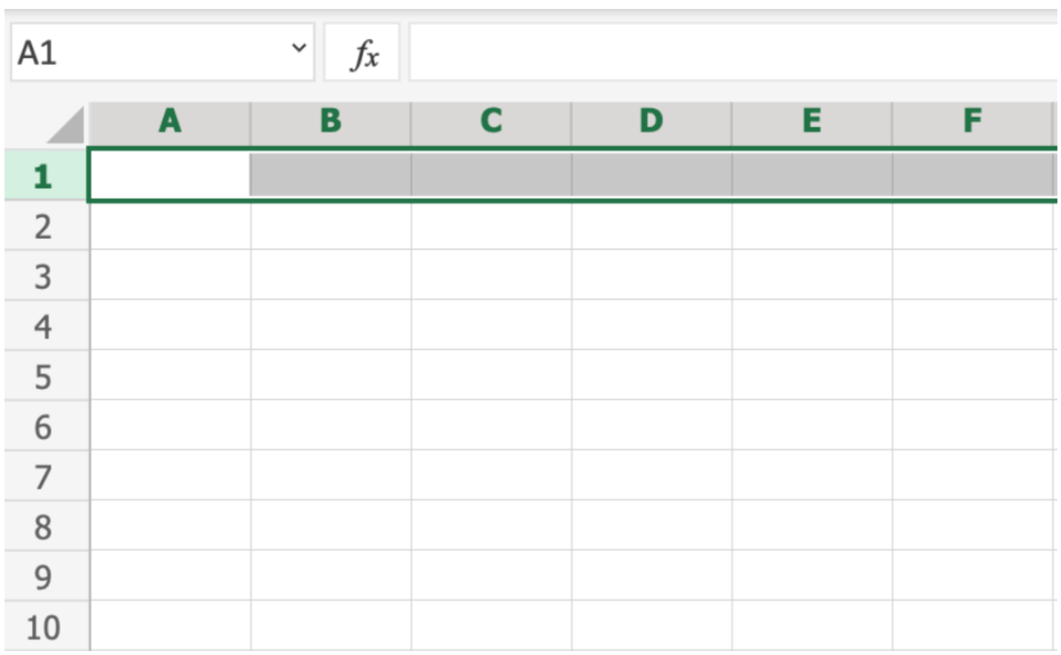
Let's try an example: Select the cells A1, A7, C1, C7 and B4



Selecting a Column: To select **column A**, click on the letter A in the column bar:

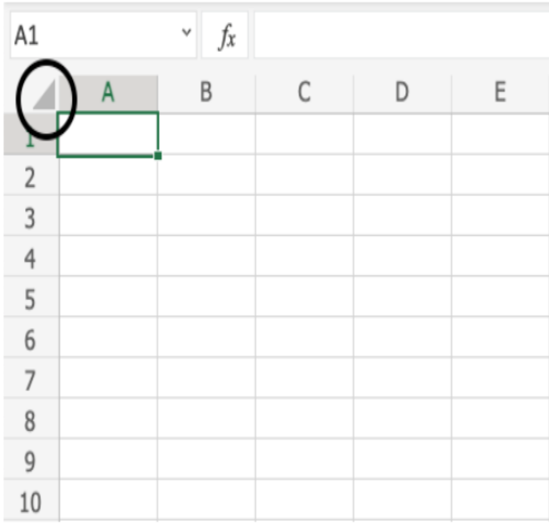


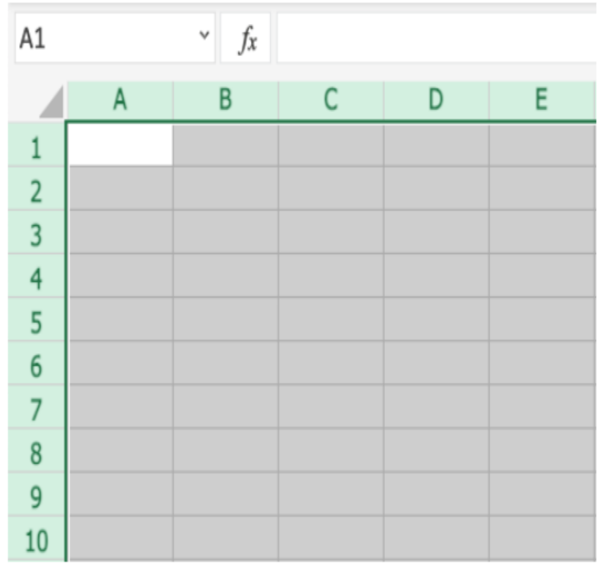
Selecting a Row: To select **row 1**, click on its number in the row bar:



What if we want to select group of rows and columns ?!

Selecting the Entire Sheet:

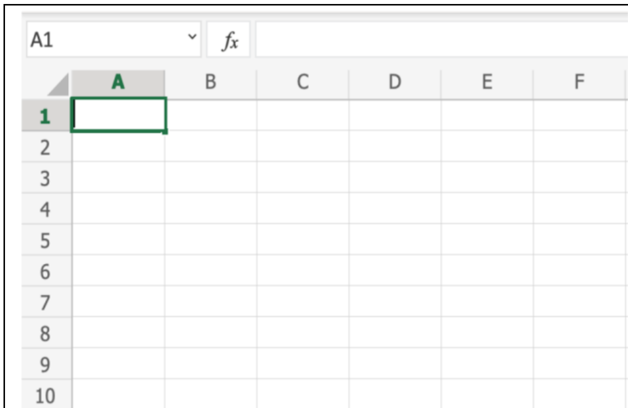


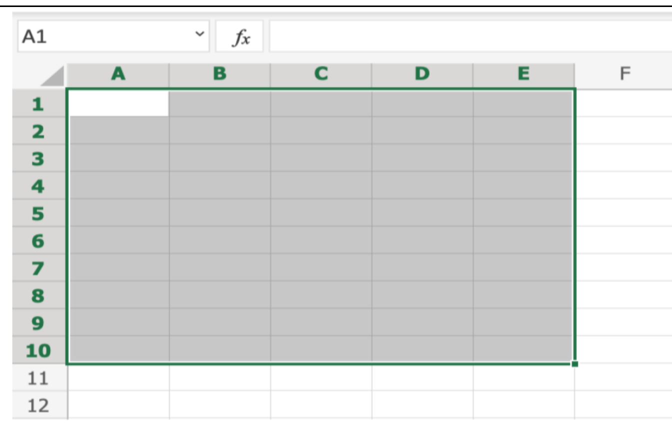


You can also select the entire spreadsheet by pressing Ctrl+A for Windows, or Cmd+A for MacOS.

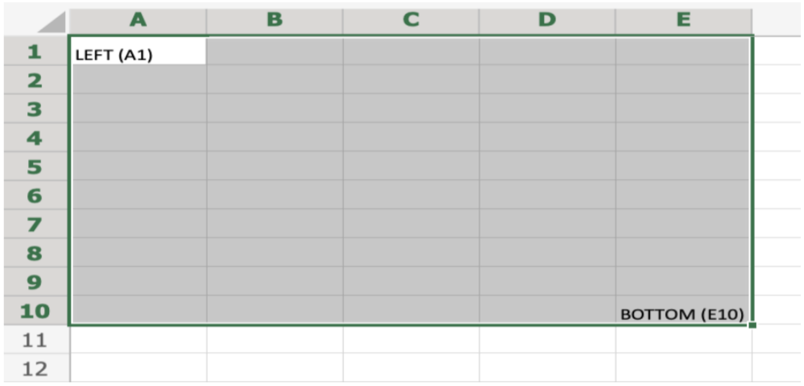
Selection of Ranges

We can do it by mouse click (Left) with drag and drop. (A1:E10)





The second way is Name Box



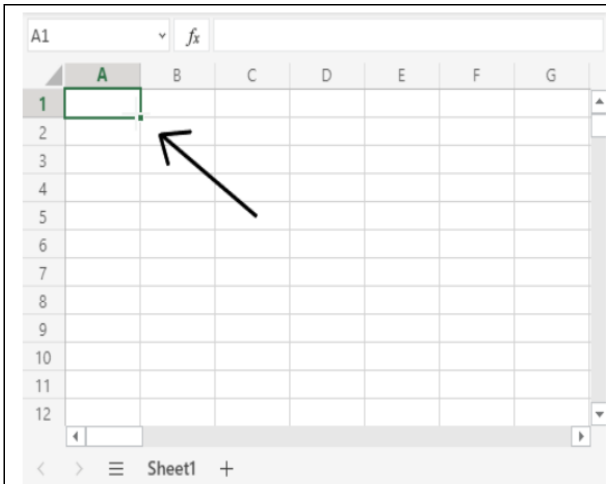
Filling

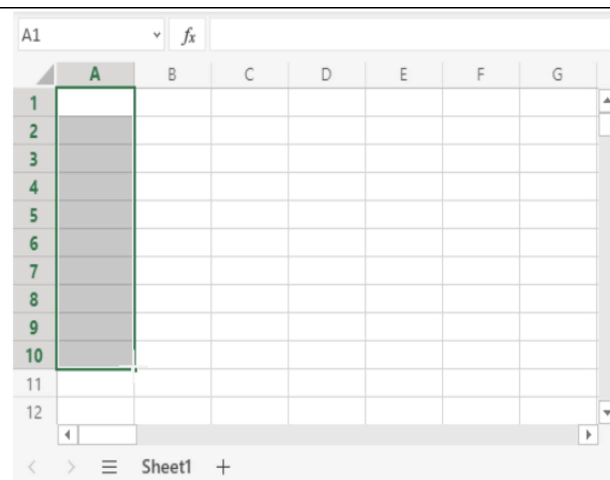
Filling makes your life easier and is used to fill ranges with values, so that you do not have to type manual entries.

Filling can be used for:

- Copying
- Sequences
- Dates
- Functions (*)

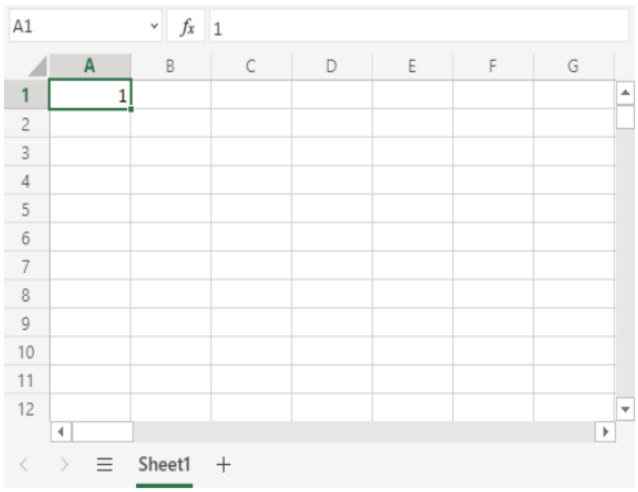
For now, do not think of functions. We will cover that in a later chapter.

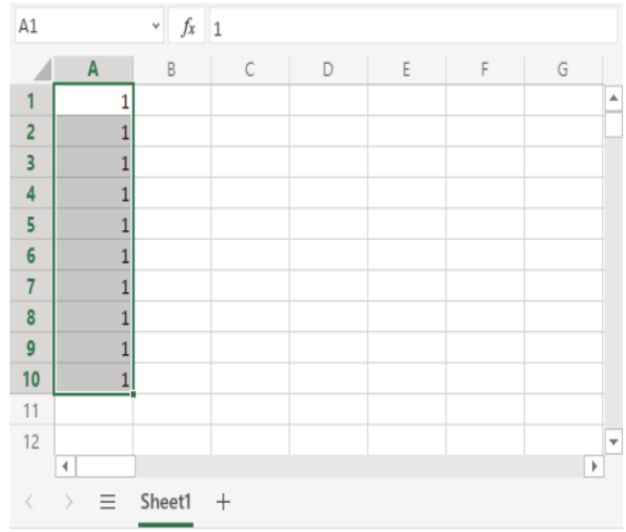




In this example, cell A1 was selected and the range A1:A10 was marked.

Fill Copies : Filling the range A1:A10 creates ten copies of 1:

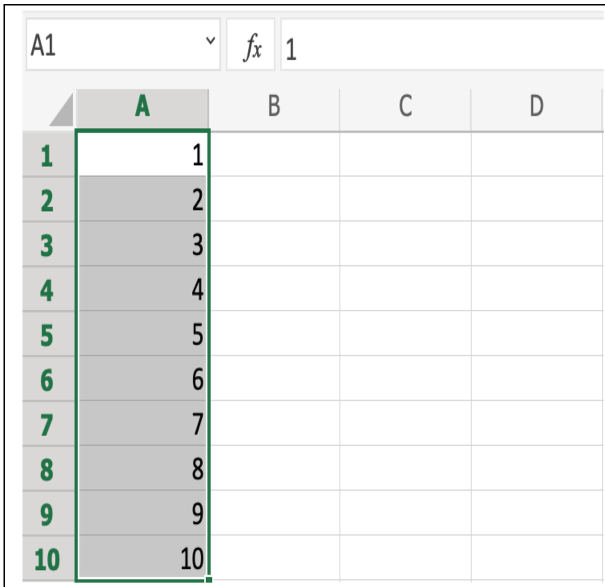


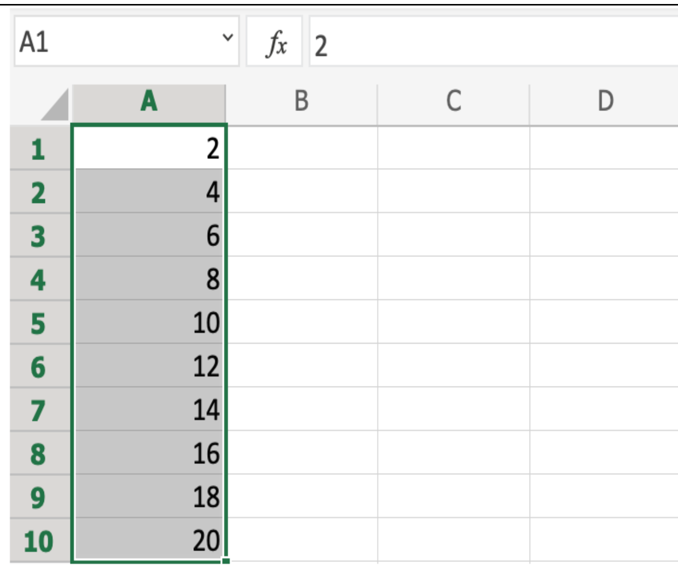


To do : In the same way type string (Hello World).

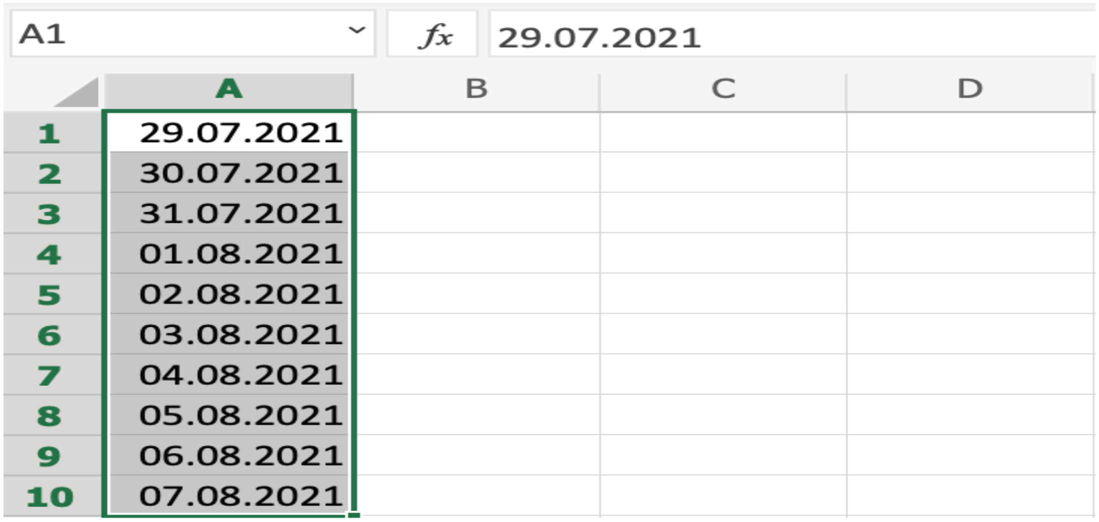
Fill Sequences

- Filling can be used to create sequences. A sequence is an order or a pattern
- Sequences can for example be used on numbers and dates.
- Let's start with learning how to count from 1 to 10.
- Example on the left we press Ctrl+ Drag and Drop.
- On the right example press Shift + drag and drop.

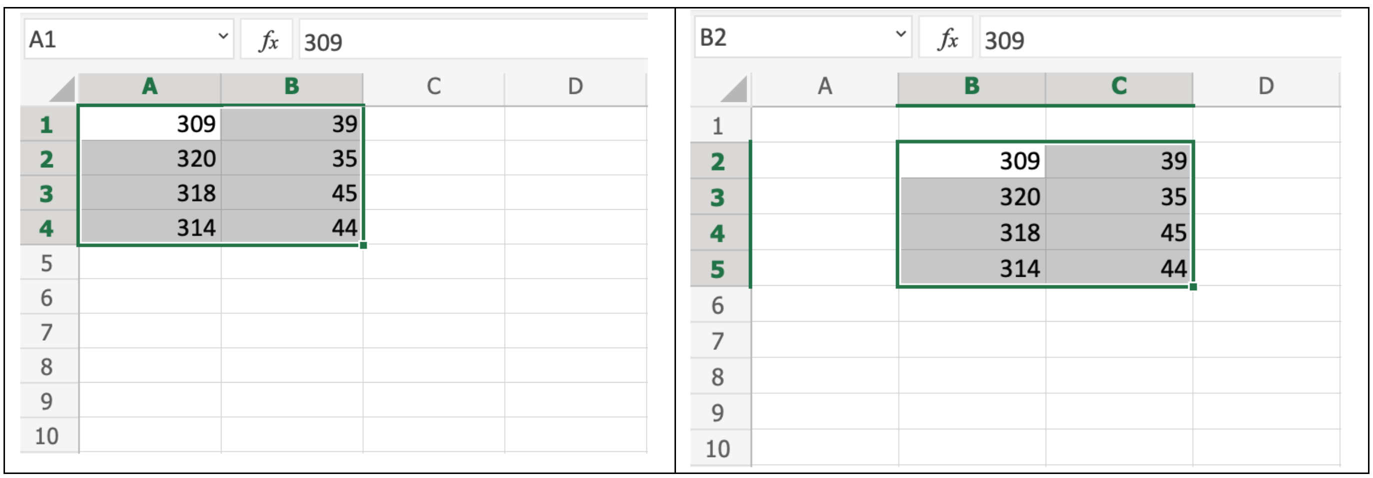




Sequence of Dates: The date format depends on you regional language settings.
For example 14.03.2023 vs. 3/14/2023 (To do lator Home Assignement)



Excel Move Cells : two ways to move cells: **Drag and drop** or by **copy and paste**.



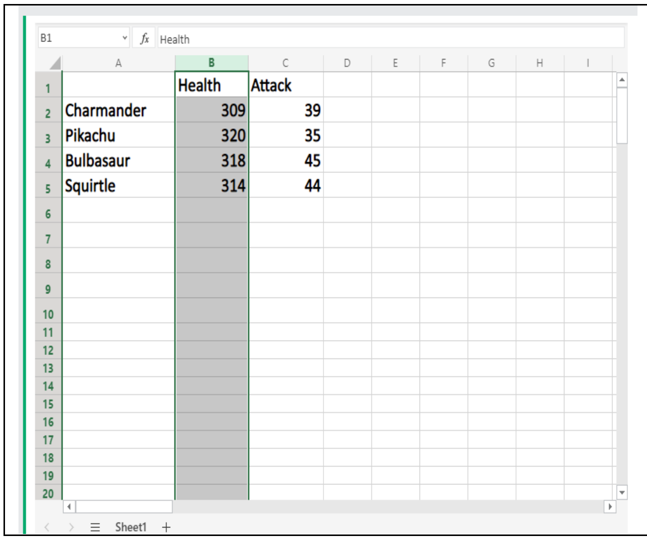
Drag and Drop(HomeWork)

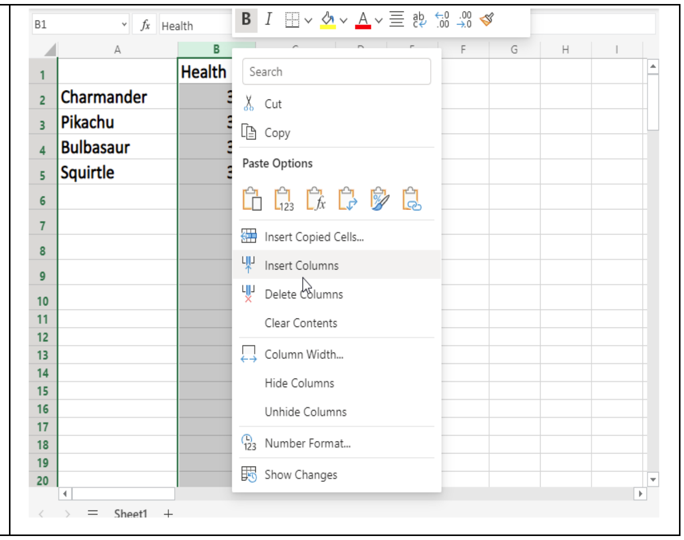
drag and drop the range by pressing and holding the left mouse button on the border. The mouse cursor will change to the move symbol when you hover over the border.

Excel Add Cells:

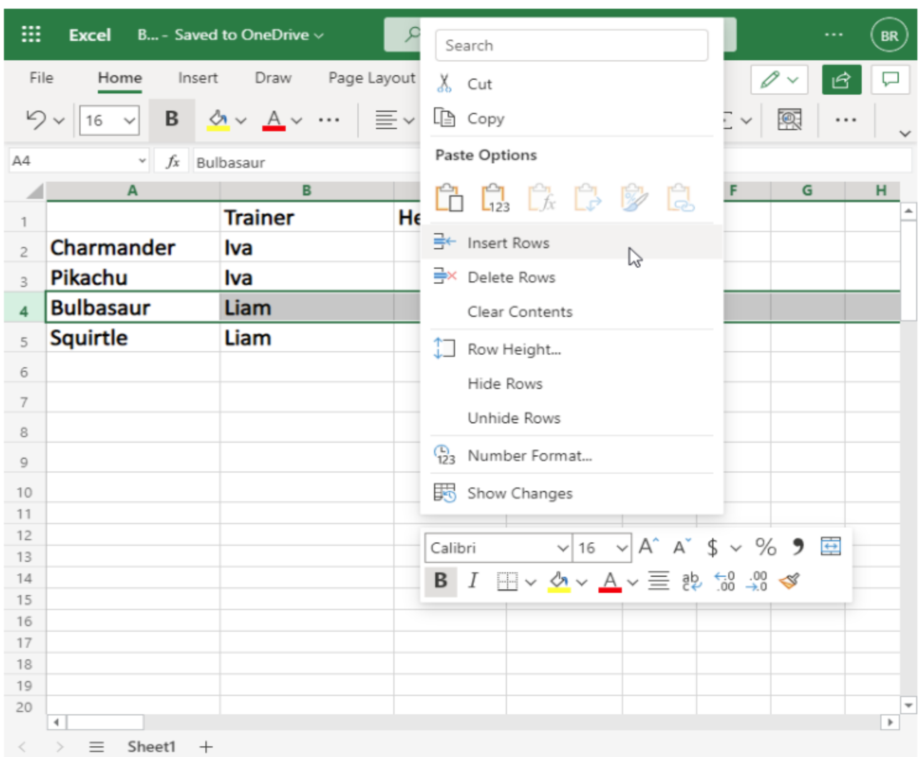
Adding New Columns (B)

Columns can be added and deleted. You access the menu by right clicking the column letter. New columns are added to the same place you clicked





Adding New Rows



Delete Cell, column and row

Undo and Redo

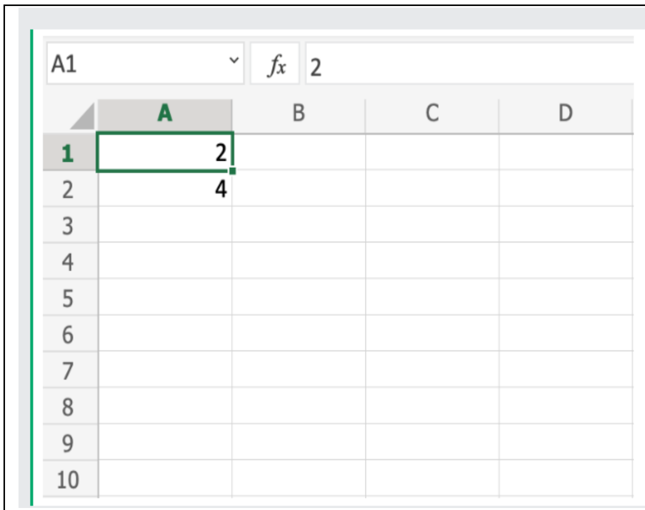
Undo is helpful if you regret an action and want to go back to how it was before.

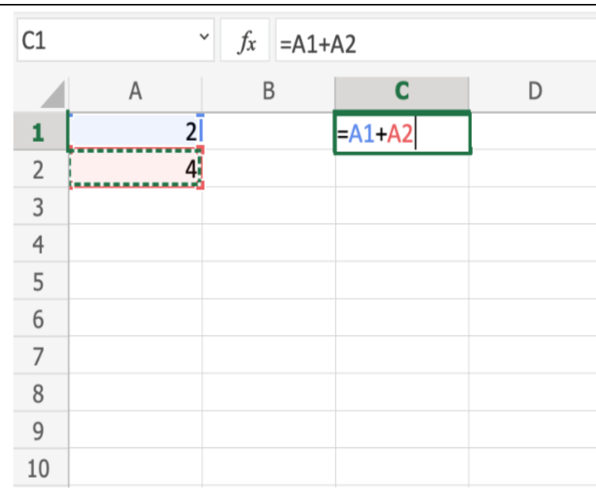
Examples of use

- Undo deleting a formula
- Undo adding a column
- Undo removing a row

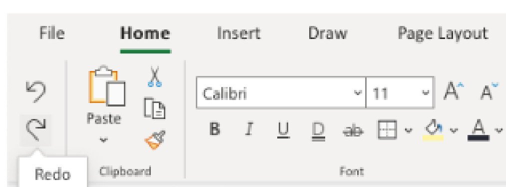
Excel Formual
Step by step:

1. Select C1 and type (=)
2. Left click A1
3. Type (+)
4. Left click A2
5. Press enter





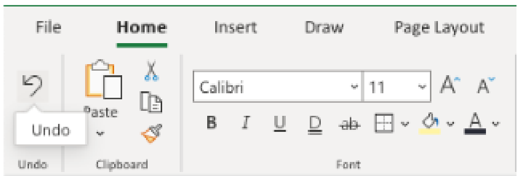
1) Pressing the Redo button in the Ribbon:



2) Using the keyboard shortcut CTRL + Y / Command + Y

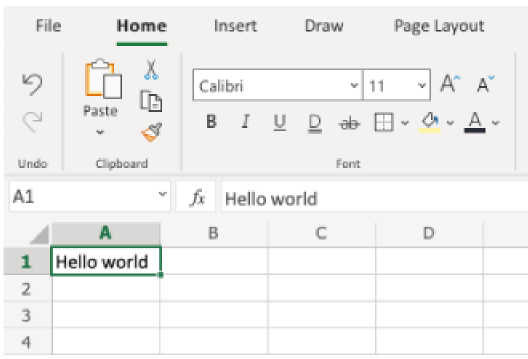
There are two ways to access the Undo command.

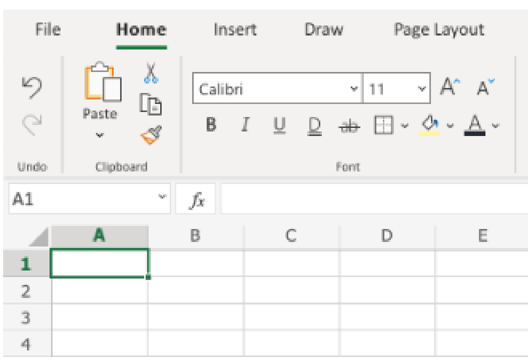
1) Pressing the Undo button in the Ribbon:



2) Using the keyboard shortcut CTRL + Z / Command + Z

Let's have a look at an example:



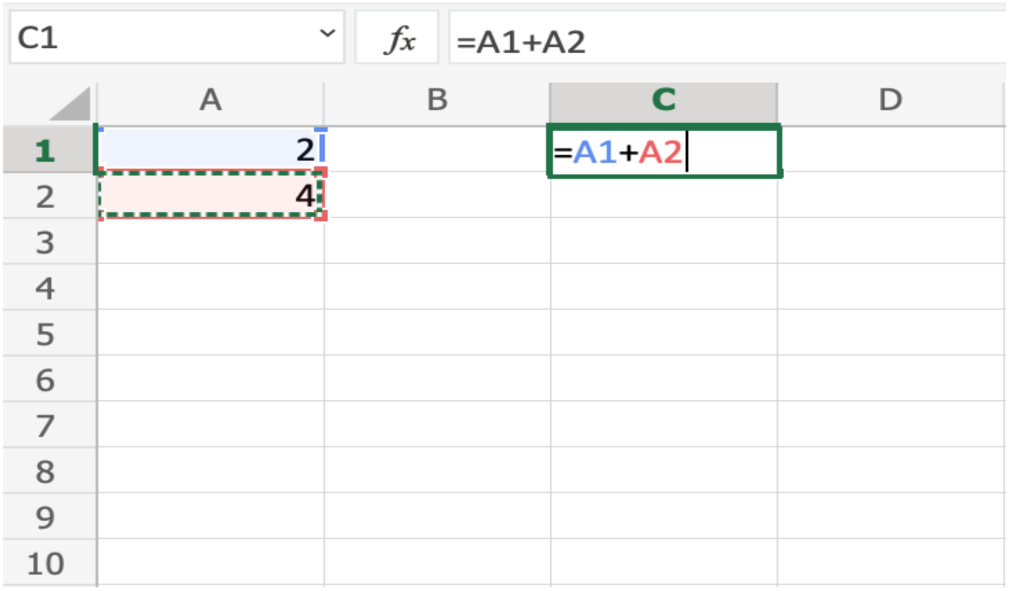


Excel Formulas

A formula in Excel is used to do mathematical calculations. Formulas always start with the equal sign (=) typed in the cell, followed by your calculation

Formulas can be used for calculations such as:

- =1+1
- =2*2
- =4/2=2



Excel Keyboard Shortcut

Description	Windows	Mac OS
Copy	Ctrl + C	Command + C or Ctrl + C
Paste	Ctrl + V	Command + V or Ctrl + V
Cut	Ctrl + X	Command + X or Ctrl + X
Undo	Ctrl + Z	Command + Z or Ctrl + Z
Redo	Ctrl + Y	Command + Y or Ctrl + Y or Command + Shift + Z
Remove Cell Contents	Delete	Delete
Bold	Ctrl + B	Command + B or Ctrl + B
Italic	Ctrl + I	Command + I or Ctrl + I
Underline	Ctrl + U	Command + U or Ctrl + U
Fill cells down	Ctrl + D	Command + D or Ctrl + D
Fill cells right	Ctrl + R	Command + R or Ctrl + R
Insert cells	Ctrl + Shift + Plus sign (+)	Ctrl + Shift + Equal sign (=)
Delete cells	Ctrl + Hyphen (-)	Command + Hyphen (-) or Ctrl + Hyphen (-)
Save	Ctrl + S	Command + S or Ctrl + S
Save As	F12	Command + Shift + S or F12
Print	Ctrl + P	Command + P or Ctrl + P
Close Window	Ctrl + W or Ctrl + F4	Command + W or Ctrl + W
Quit Excel	Alt + F4	Command + Q

Ref:

- https://support.microsoft.com/ar-sa/office/%D8%A7%D9%84%D9%85%D9%87%D8%A7%D9%85-%D8%A7%D9%84%D8%A3%D8%B3%D8%A7%D8%B3%D9%8A%D8%A9-%D9%81%D9%8A-excel-dc775dd1-fa52-430f-9c3c-d998d1735fca
- https://www.datacamp.com/courses/introduction-to-excel
- https://www.datacamp.com/courses/data-analysis-in-excel
- https://www.datacamp.com/courses/data-preparation-in-excel
- https://www.datacamp.com/courses/data-visualization-in-excel

Exercise

1- Type the value that the formula returns: =2*15

2- Complete the sequence of dates:

A1(07.20.2025)

A2(07.21.2025)

A3()

3- Type the value that the SUM function returns: =SUM(25;15;5)

4- Type the value that the formula returns: =(2*(2+2)+(4+4)*2):

5- Type the range **Column A, Row 2 : Column B, Row 5**:

6- Complete the sequence of numbers:

A1(2)

A2(4)

A3

7- Type reference A1 as **absolute**: …… A$1